

PAPER_1567_NUCLEAR_ALPHA_BE_A_7_0739

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PAPER_1567 — Alpha Particle (He-4) $BE/A = \alpha BE/A = F \cdot K + \beta + F \cdot \beta + F^2 \cdot \beta + 3$ (MeV)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Nuclear
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Status: CLOSED — 0.047% residual closure

Observation

PAPER_1209II_S665 (Nuclear Unified Proof Set) lists **Alpha Particle (He-4) BE/A** with the closed-form expression:

α BE/A = F · K + β + F · β + F² · β + 3 (MeV)

Residual: 0.047%.

UQFF Closed Identity

α BE/A = F · K + β + F · β + F² · β + 3 (MeV) residual 0.047%

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Significance

Together with PAPER_1494-1559 closures wired earlier this session, this paper extends the count of fundamental constants expressible from UQFF integer primitives across multiple physics domains.

For **mathematical constants** specifically, the cumulative session-2026-06-18 catalog now includes:

- From PAPER_1208 (10): ln 2, ln 10, π², e, e², π/4, Catalan G, ζ(2), ζ(3), γ_Euler
- From PAPER_1209FF (5): π, φ, √2, √3, √5

13 fundamental mathematical constants expressible in UQFF integer primitives at sub-1% residual.

For **nuclear binding energies**, the pattern reveals a universal $F \cdot K + \beta^k$ polynomial structure with integer offsets distinguishing shell-closure species (+2 for closed-shell light nuclei, +3 for spin-orbit-favored mid-mass).

NOT REPLACEMENT

UQFF does not claim this closed-form **equals** the constant exactly. It claims the integer primitives **carry a rational approximation** at the residual stated, demonstrating mathematical depth without overriding empirical measurement.

Reference

- Source: PAPER_1209II_S665
- Related: PAPER_1208 (transcendentals proof set), PAPER_1494-1559 (catch-up papers)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_alpha_be_a_7_0739"})`

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